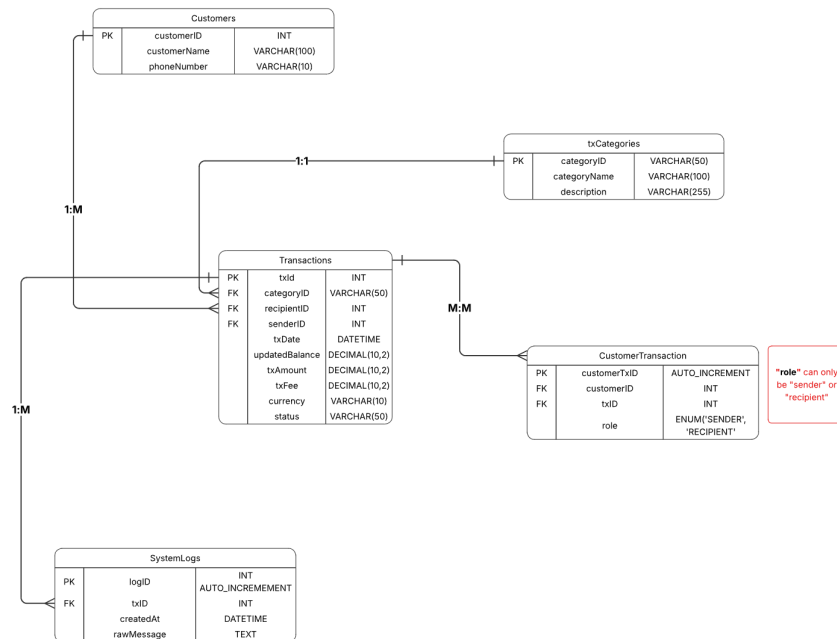


DATABASE DESIGN DOCUMENT

Entity-Relationship Design Diagram And Documentation



Design Rationale and Justification

Our ERD provides a structured way to store, organize, and manage Mobile Money transaction information extracted from SMS/XML data. It includes five main entities: Transactions, Customers, txCategories, SystemLogs, and CustomerTransaction, as shown in the diagram above.

The Transactions table is the central entity because it contains the main information about each MoMo transaction. Its attributes include transaction amounts, fees, dates, times, currency, status, and the sender and recipient relationships. The Transactions table is connected to the Customers table through the sender and recipient information. The Customers table stores customer details separately so that the same customer can be associated with multiple transactions without their personal information being repeatedly stored.

The txCategories table was introduced to classify transactions according to their type. We use a separate category table to add new transaction categories without changing the structure of the Transactions table. The category ID is used as a foreign key in Transactions to connect each transaction to its category. This is a one-to-many relationship,

because one transaction category can be associated with many transactions, while each transaction is assigned to a category.

We also created the CustomerTransaction junction table. It connects customers to transactions using customerID and txID. The role attribute further explains the relationship by showing whether the customer was the SENDER or RECIPIENT. This makes the relationship more meaningful and helps the database distinguish the different roles a customer can have in a transaction.

The SystemLogs table is connected to the Transactions table to keep track of transaction processing and related system information. It stores the original raw message, the time the log was created, and the transaction it is associated with. This relationship allows the system to trace a transaction back to its original message when needed, which can be useful for monitoring and troubleshooting.

Data Dictionary

Customers

Stores the basic information of customers who participate in mobile money transactions.

Column	Data Type	Key	Description
customerID	INT	PK	Uniquely identifies the customer
CustomerName	VARCHAR (100)	–	Stores the customer name
phoneNumber	VARCHAR (10)	–	Stores the customer's mobile phone number

Transactions

Stores the main details of each mobile money transaction processed by the system.

Column	Data Type	Key	Description
txId	INT	PK	Unique identifier for a transaction
categoryID	VARCHAR (50)	FK	Identifies the category/type of transaction
recepientID	INT	FK	Identifies the customer receiving the transaction
senderID	INT	FK	Identifies the customer sending the transaction

txDate	DATETIME		Records when the transaction occurred
updatedBalance	DECIMAL (10,2)		Stores the customer's balance after the transaction has been processed.
txAmount	DECIMAL (10,2)		Stores the amount of money involved in the transaction.
txFee	DECIMAL (10,2)		Stores the fee charged for processing the transaction.
currency	VARCHAR (10)		Stores the currency used for the transaction, which is RWF.
status	VARCHAR (50)		Records the current status of the transaction, such as successful, pending, or failed.

Transaction Categories

Stores the different categories or types of mobile money transactions supported by the system.

Column	Data Type	Key	Description
categoryID	VARCHAR (50)	PK	A unique identifier assigned to each transaction category.
categoryName	VARCHAR (100)		Stores the name of the transaction category.
description	VARCHAR (255)		Provides additional information about what the transaction category represents.

System Logs

Stores records of events and messages generated while the system processes mobile money transactions. This can help with monitoring, troubleshooting, and tracking processing activities.

Column	Data Type	Key	Description
logID	INT	PK	A unique identification number assigned to each system log entry.
txID	INT	FK	Identifies the transaction associated with the system log.
createdAt	DATETIME		Records the date and time when the log entry was created.
rawMessage	TEXT		Stores the original or raw message received from the mobile money data source.

Customers Transactions

Acts as a junction table connecting customers to transactions. It allows a transaction to be associated with multiple customers while also identifying the role each customer played.

Column	Data Type	Key	Description
customerTxID	AUTO_INCREMENT	PK	A unique identification number assigned to each customer-transaction relationship.
customerID	INT	FK	Identifies the customer associated with the transaction.
txID	INT	FK	Identifies the transaction associated with the customer.
role	ENUM ('SENDER',	–	Identifies whether

	'RECIPIENT')		the customer was the sender or recipient in the transaction.
--	--------------	--	--

Sample Queries and Tests

```

54
55 • USE momosmsDB;
56
57 • INSERT INTO txCategories (categoryID, categoryName, description)
58   VALUES ('DEP', 'Deposit', 'Funds added to account');
59
60 • INSERT INTO Customers (customerName, phoneNumber)
61   VALUES ('Oscar Isaac', '0759674567');
62
63 • INSERT INTO Customers (customerName, phoneNumber)
64   VALUES ('Karyna Kirabo', '0744089764');
65
66
67

```

Output
OK, 0 records retrieved in 3.143ms

- Adding new customers

```

58 -- SAMPLE QUERIES --
59
60 • INSERT INTO txCategories (categoryID, categoryName, description)
61   VALUES ('DEP', 'Deposit', 'Funds added to account');
62
63 • INSERT INTO customers (customerName, phoneNumber)
64   VALUES ('Oscar Isaac', '0759674567');
65
66 • INSERT INTO customers (customerName, phoneNumber)
67   VALUES ('Karyna Kirabo', '0744089764');
68
69 ✓ • INSERT INTO transactions (
70   | categoryID, recipientID, senderID, txDate,
71   | updatedBalance, txAmount, txFee, currency, status
72   | )
73   VALUES (
74   | 'DEP', 2, 1, NOW(), 25000.00, 5000.00, 100.00, 'RWF', 'SUCCESS'
75   | );
76
77
78 • INSERT INTO customerTransaction (customerID, txID, role)
79   VALUES (1, 1, 'sender');
80
81 • INSERT INTO customerTransaction (customerID, txID, role)
82   VALUES (2, 1, 'recipient');
83
84
85 • SELECT t.txID, s.customerName AS sender, r.customerName AS recipient, t.txAmount, t.currency, t.status
86   FROM transactions t
87   JOIN customers s ON t.senderID = s.customerID
88   JOIN customers r ON t.recipientID = r.customerID;
89

```

Output **Result #1**

txID	sender	recipient	txAmount	currency	status
1	Oscar Isaac	Karyna Kirabo	5000.00	RWF	SUCCESS

- Creating transactions

```

96
97     -- full transaction via junction TABLE
98
99     • SELECT cu.customerName, ct.role, t.txId, t.txAmount
100     FROM customerTransaction ct
101     JOIN customers cu ON ct.customerID = cu.customerID
102     JOIN transactions t ON ct.txID = t.txID
103     WHERE cu.customerID = 1;
104

```

Output	Result #1			
customerName	role	txId	txAmount	
Oscar Isaac	sender	1	5000.00	

- Joining customer name and transaction details

```

105
106     -- test rules
107     -- 1. Sender cannot send to themselves
108     • INSERT INTO transactions (
109         categoryID, recipientID, senderID, txDate,
110         updatedBalance, txAmount, txFee, currency, status
111     )
112     VALUES (
113         'DEP', 1, 1, NOW(), 25000.00, 5000.00, 100.00, 'RWF', 'SUCCESS'
114     );
115

```

Output

MySQL Error (3819): Check constraint 'transactions_chk_5' is violated.

- Testing if the sender can send money to themselves.

```

116
117     -- 2. Cannot have negative amount transacted
118
119     • INSERT INTO transactions(      categoryID, recipientID, senderID, txDate,
120         updatedBalance, txAmount, txFee, currency, status
121     )
122     VALUES (
123         'DEP', 1, 1, NOW(), -25000.00, 5000.00, 100.00, 'RWF', 'SUCCESS'
124     );
125

```

Output

MySQL Error (3819): Check constraint 'transactions_chk_5' is violated.

- Testing if one can have a -ve transaction